WORKED SOLUTION

STEP II 1991, Question 3

Pure • Symmetric Simultaneous Equations

STEP II 1991, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q3

The key idea

Three equal expressions give two equations, and with three unknowns we can't hope to find x, y, z themselves. What we can do is exploit the cyclic symmetry. Subtracting pairs of expressions turns each difference $x - y$ into a multiple of the next difference $y - z$, and multiplying the three relations together makes almost everything cancel. For the second result, giving the common value a name k and clearing fractions produces three cyclic linear relations that pin down xyz in three different ways.

Solution

Showing that $x^2y^2z^2 = 1$

Equating the first two expressions,

$$x - y = \frac{1}{z} - \frac{1}{y} = \frac{y - z}{yz}.$$

In the same way, equating the second and third, and then the third and first,

$$y - z = \frac{1}{x} - \frac{1}{z} = \frac{z - x}{zx}, \qquad z - x = \frac{1}{y} - \frac{1}{x} = \frac{x - y}{xy}.$$

Multiplying these three equations together gives

$$(x - y)(y - z)(z - x) = \frac{(y - z)(z - x)(x - y)}{x^2y^2z^2}.$$

Because x, y and z are distinct, the product $(x - y)(y - z)(z - x)$ is not zero, so we may divide both sides by it. This leaves $1 = 1/(x^2y^2z^2)$, that is,

$$x^2y^2z^2 = 1,$$

as required.

Ask yourself Where exactly did we use that x, y, z are distinct and non-zero?

Non-zero is needed for the fractions $1/x$, $1/y$, $1/z$ to make sense at all. Distinctness is what lets us cancel $(x-y)(y-z)(z-x)$. Without it the result fails: $x=y=z=2$ satisfies the equations, with $x^2y^2z^2=64$.

The common value is ± 1

Let k be the common value, so

$$x+\frac{1}{y}=k, \qquad y+\frac{1}{z}=k, \qquad z+\frac{1}{x}=k.$$

Multiplying through by y, z and x respectively (all non-zero),

$$xy=ky-1, \qquad yz=kz-1, \qquad zx=kx-1. \tag{$*$}$$

Now multiply the first of these by z and use the second to replace yz:

$$\begin{aligned} xyz &= k\,yz - z \\ &= k(kz-1)-z \\ &= (k^2-1)z-k. \end{aligned}$$

The relations $(*)$ are unchanged when we replace $x \to y \to z \to x$, so the same calculation, started from the second and third relations, gives

$$xyz=(k^2-1)x-k, \qquad xyz=(k^2-1)y-k.$$

Subtracting the first two expressions for xyz,

$$(k^2-1)(z-x)=0.$$

Since $z \neq x$, we must have $k^2=1$, so the value of $x+\frac{1}{y}$ is either $+1$ or -1, as required.

As a bonus, putting $k^2=1$ back in gives $xyz=-k$. This is a second route to the first result, since then $x^2y^2z^2=k^2=1$.

Checks

Both values really occur. With $k=1$, start from any $x \neq 0,1$ and set $y=\frac{1}{1-x}$ and $z=\frac{1}{1-y}=\frac{x-1}{x}$. Then $z+\frac{1}{x}=\frac{x-1}{x}+\frac{1}{x}=1$, so all three expressions equal 1. For instance $x=2$ gives $y=-1$, $z=\frac{1}{2}$: indeed $2-1=-1+2=\frac{1}{2}+\frac{1}{2}=1$, and $xyz=-1=-k$. Replacing (x,y,z) by $(-x,-y,-z)$ gives solutions with $k=-1$, such as $(-2,1,-\frac{1}{2})$. The three numbers are always distinct here, because $x=\frac{1}{1-x}$ would need $x^2-x+1=0$, which has no real roots. We checked about 4000 such triples with exact fractions, confirming each time that $x^2y^2z^2=1$ and $xyz=-k$.

Common slips

- Dividing by $(x-y)(y-z)(z-x)$ without saying why it is non-zero. The distinctness condition is there precisely for this.
- Concluding $xyz = 1$ instead of $xyz = \pm 1$: in fact $xyz = -k$, so $xyz = -1$ when the common value is 1.

Beyond STEP

The map $t \mapsto \dfrac{1}{k-t}$ takes x to y, y to z and z back to x. It is a Möbius transformation, with matrix $\begin{pmatrix} 0 & 1 \\ -1 & k \end{pmatrix}$, and composing such maps corresponds to multiplying their matrices. The question is really asking when this map has a 3-cycle, and the answer, $k^2 = 1$, is exactly when the cube of the matrix is a multiple of the identity. Needham's *Visual Complex Analysis* has a beautiful chapter on Möbius transformations and their classification by the trace of the matrix.

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